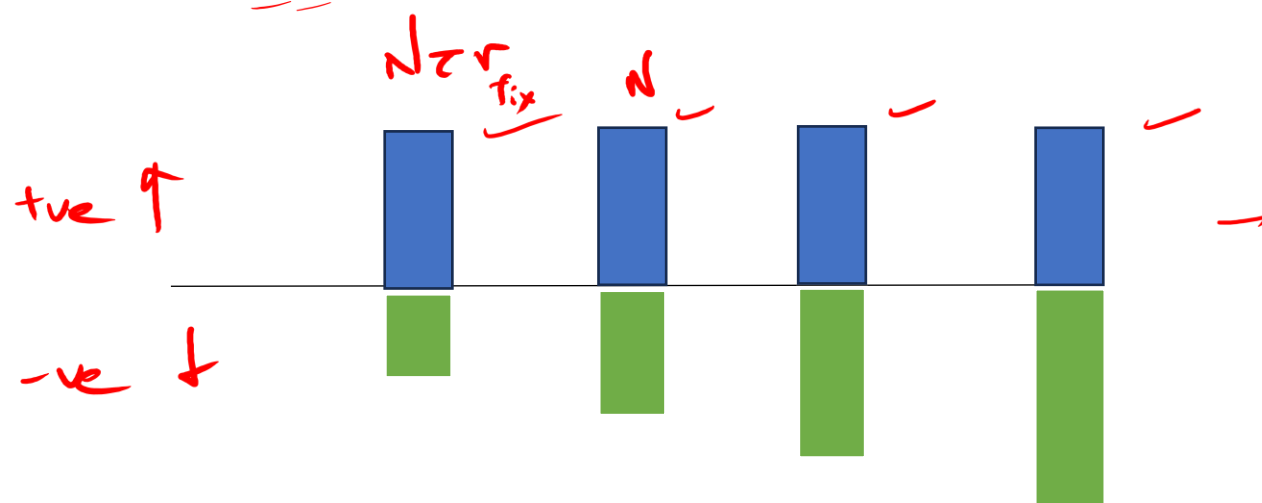


RFR Regime (3. SOFR Swaps)

-

Fixed-Floating Swap Basics

- ▶ One party pays fixed coupons whereas another party pays floating coupons tied to a benchmark
- ▶ The floating payment may or may not include an additional spread
- ▶ The fixed rate that makes the value of the swap = 0 on any day is the swap rate or par rate.
Par rates can be considered as average of the floating rates over the life of the instrument.
- ▶ Long on swap = long on floating leg = fixed payer = payer swap ✓
- ▶ Short on swap = short on floating leg = fixed receiver = receiver swap ✓

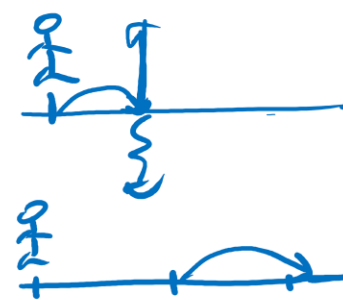


(typical cashflow profile for a receiver swap under an upward sloping yield curve)

USD SOFR Swap Template

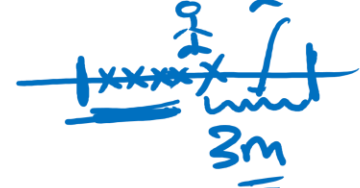
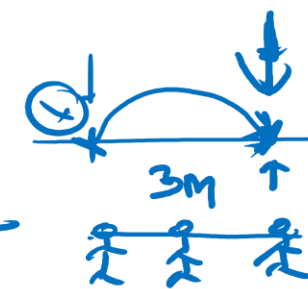
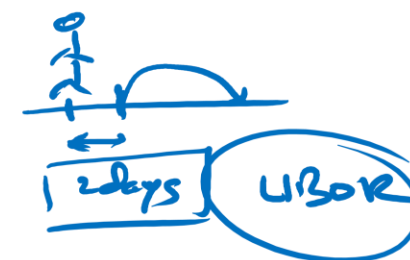
Swap	Leg1 : FIXED	Leg2 : FLOAT
Pay/Receive	RECEIVE	PAY
Notional	100,000,000	100,000,000 →
Currency	USD	USD
Effective Date	0D Thu, 20-March-2025	0D Thu, 20-March-2025
Termination Date	1Y Fri, 20-March-2026	1Y Fri, 20-March-2026
Fixed Rate (%)	3.4%	
Float Index		1D USD SOFR
Float Spread (bps)		0.00 +
Reset Frequency		DAILY
Payment Frequency	SEMI-ANNUALLY	QUARTERLY
Day Count Convention	30/360	ACT/360

Market	Leg1 : FIXED	Leg2 : FLOAT
Curve Date	0D Thu, 20-March-2025	0D Thu, 20-March-2025
Forecast Curve		USD SOFR → 3M LIBOR
Discount Curve	USD SOFR	USD SOFR



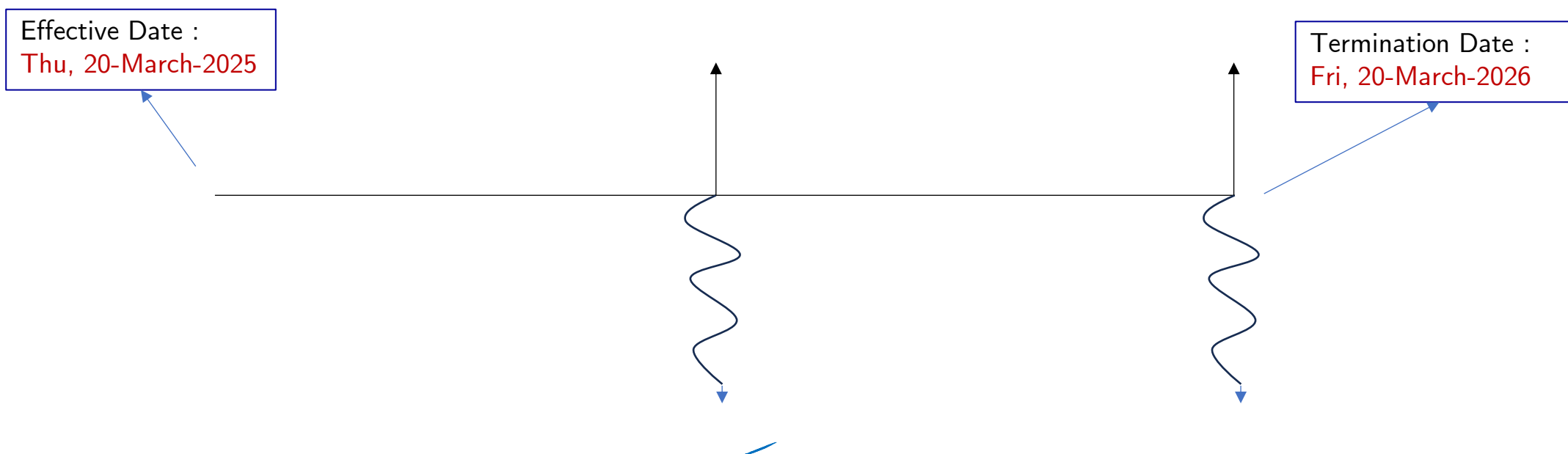
$$N \times T \times r$$

$$T(d_1, d_2)$$



$$\left[\prod \left(1 + r_i \cdot \frac{d_i}{360} \right) - 1 \right] \frac{360}{d_i}$$

Effective Date & Termination Date



- ▶ **Effective Date** : Start date of the swap
- ▶ **Termination Date** : End date (Maturity of the swap)

Calendar

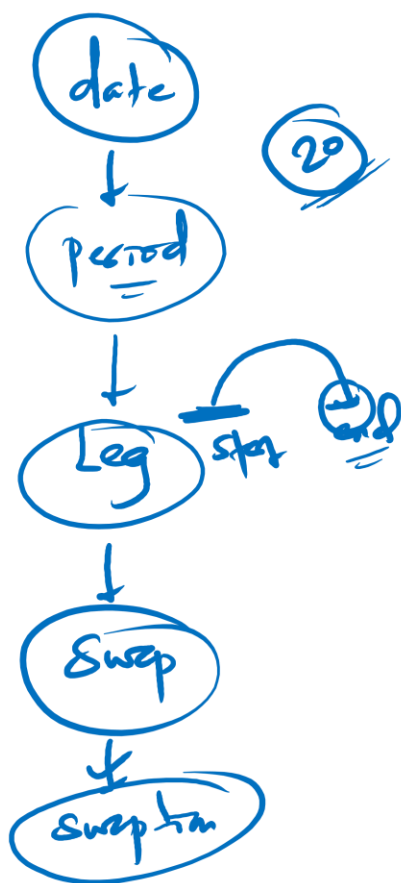


- ▶ A calendar is a set $\mathcal{C}$ of non-business days or holidays. Examples of calendars are TARGET (Trans-European Automated Real-time Gross settlement Express Transfer), LONDON, NYC etc.
- ▶ A day d is non-business day if $d \in \mathcal{C}$
- ▶ Calendars need to be specified in the context of a financial product

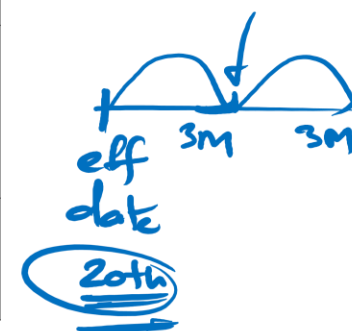


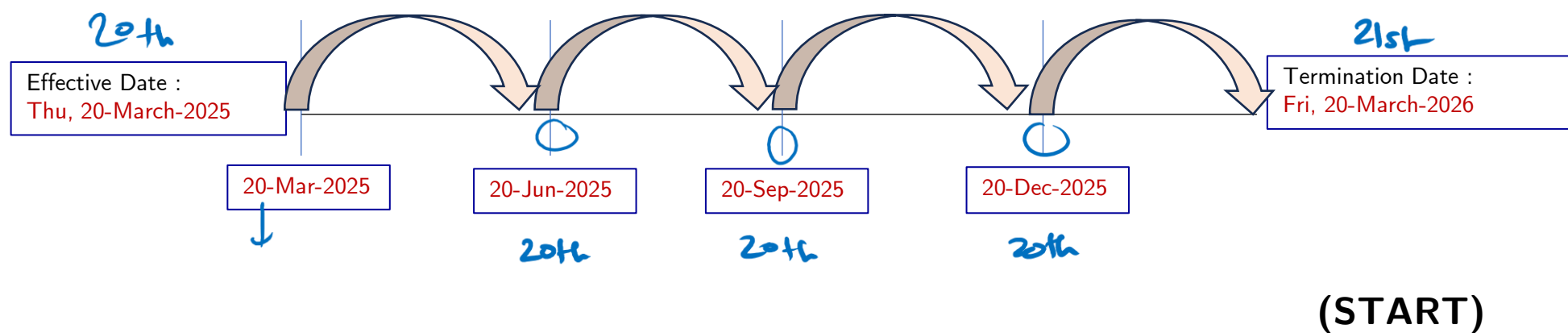
Roll Day

- **Roll day** is used to determine the unadjusted coupon accrual end dates. It can be specified either by a number between 1-31 or one of the values in the list { IMM, EOM, START, END }



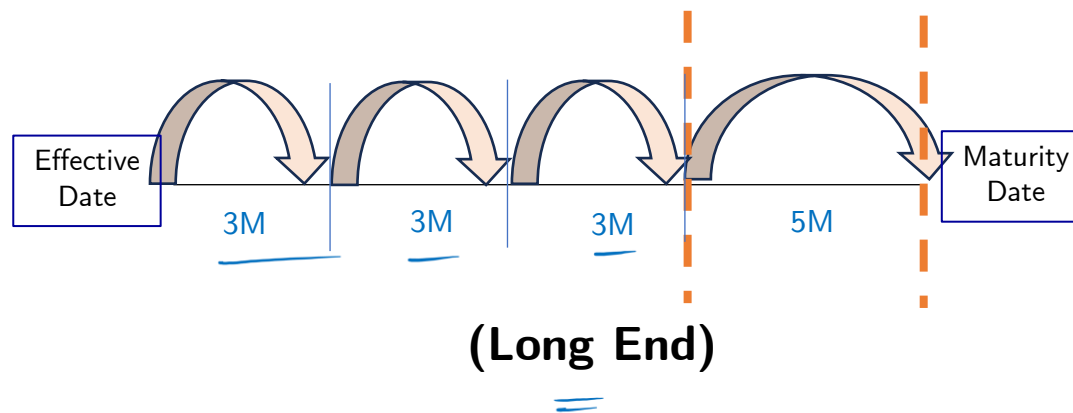
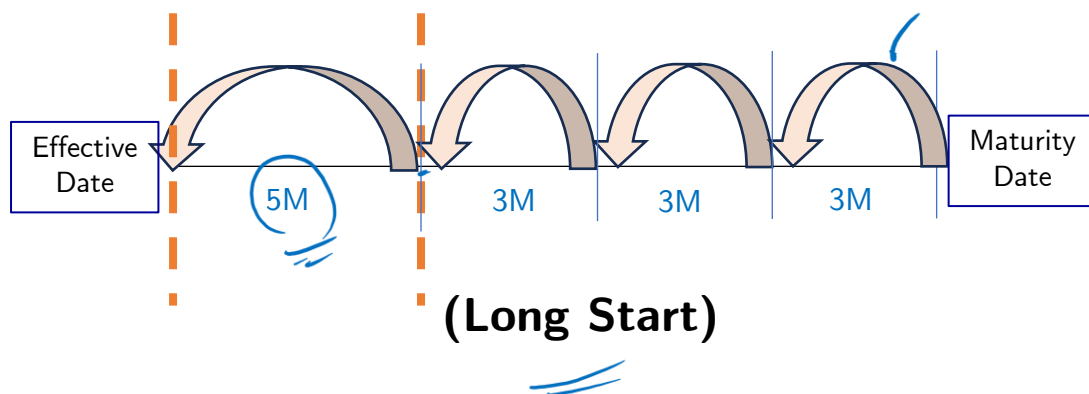
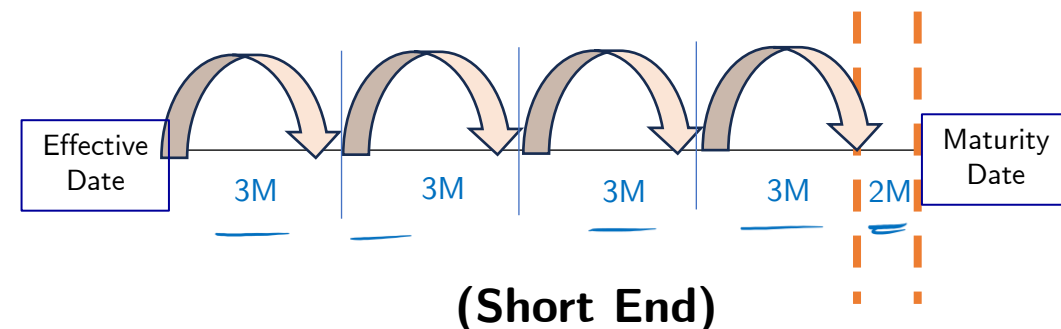
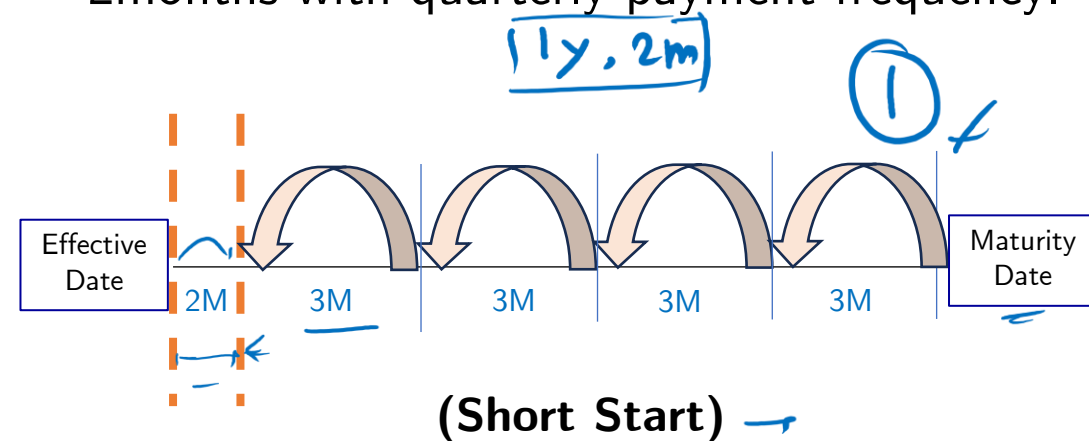
Roll Day	Method
<u>1-31</u>	The coupon accrual end date falls on this day of the month
<u>IMM</u>	Accrual end date is an IMM date i.e., 3 rd Wednesday of <u>March</u> , <u>June</u> , <u>September</u> or <u>December</u>
<u>EOM</u> <u>31st, 30, 28, 29</u>	Coupons accrue to the end of the month
<u>START</u>	Coupons accrue to the same day as the <u>unadjusted effective date</u>
<u>END (default)</u>	Coupons accrue to the same day as the <u>unadjusted termination date</u>





Stub type

- For swaps of irregular length, typically there will be one coupon period which will be shorter or longer than others which are referred to as stubs. For example, consider a swap length of 1Y 2months with quarterly payment frequency. We have the following variants



Day count convention

- Day count convention is used to calculate the year fraction between 2 dates which is used to calculate accrued coupon amount. It's a function $\tau : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$

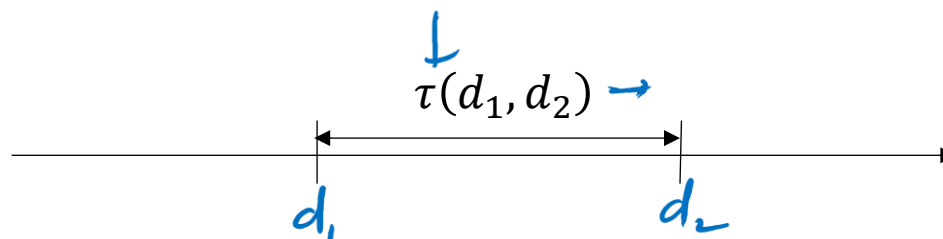
$\downarrow$
 $\mathcal{D} = \{d_i\}$
 $\begin{matrix} 0.25 \\ 0.5 \end{matrix}$

- Act / 365 : $\tau(d_1, d_2) = (d_2 - d_1) / 365 \rightarrow$

- Act / 360 : $\tau(d_1, d_2) = (d_2 - d_1) / 360$

$\mathcal{D}_i = [d_i, m_i, y_i]$

- 30 / 360 : $\tau(d_1, d_2) = \frac{(\max(30 - d_1, 0) + \min(d_2, 30) + 360 \times (y_2 - y_1) + 30 \times (m_2 - m_1))}{360}$

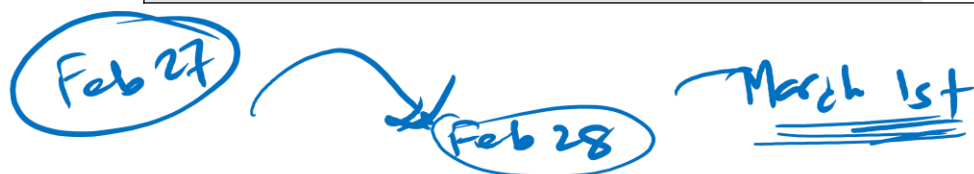


Business Day Convention

- ▶ A business day convention is a function $\omega_{\mathcal{C}}: \mathcal{D} \rightarrow \mathcal{D}$ that maps a date to another date given a calendar $\mathcal{C}$. Business days remain unchanged i.e., $\omega_{\mathcal{C}}(d) = d$ if $d \in \mathcal{C}$.
- ▶ It's a means to adjust the accrual/payment days to business days if they fall on a holiday. We have the following conventions

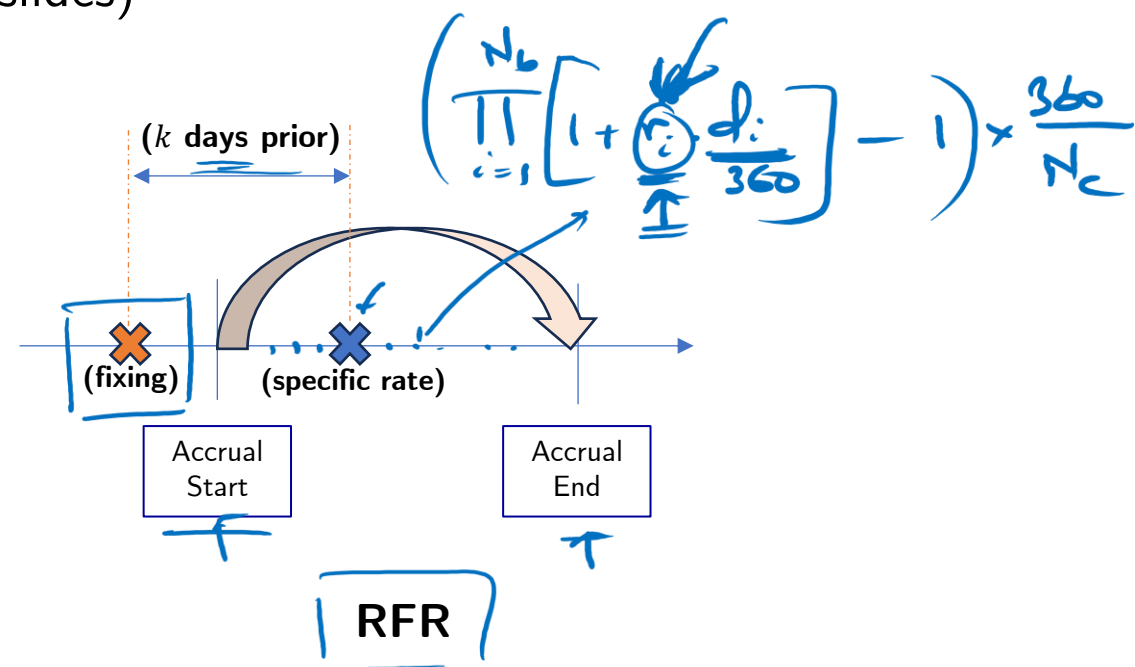
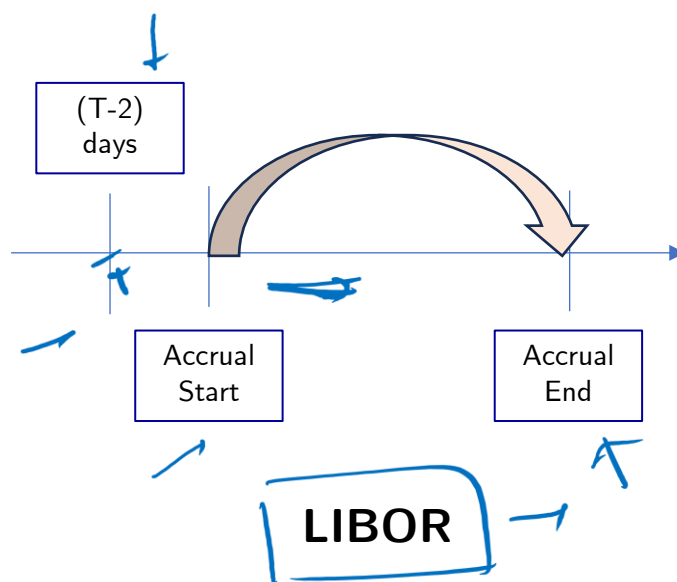


Business Day Convention	Method
<u>FOLLOWING</u> →	Adjust to the next business day $\omega_{\mathcal{C}}(d) = \min \{ \tilde{d} \tilde{d} \notin \mathcal{C}, \tilde{d} \geq d \}$
<u>PRECEDING</u> →	Adjust to the previous business day $\omega_{\mathcal{C}}(d) = \max \{ \tilde{d} \tilde{d} \notin \mathcal{C}, \tilde{d} \leq d \}$
<u>MODIFIED FOLLOWING</u> <u>default</u>	Adjust to next business day, unless it falls on the next month in which case adjust to the previous business day
<u>MODIFIED PRECEDING</u>	Adjust to previous business day, unless it falls on the previous month in which case adjust to the next business day
<u>UNADJUSTED</u> →	No Change



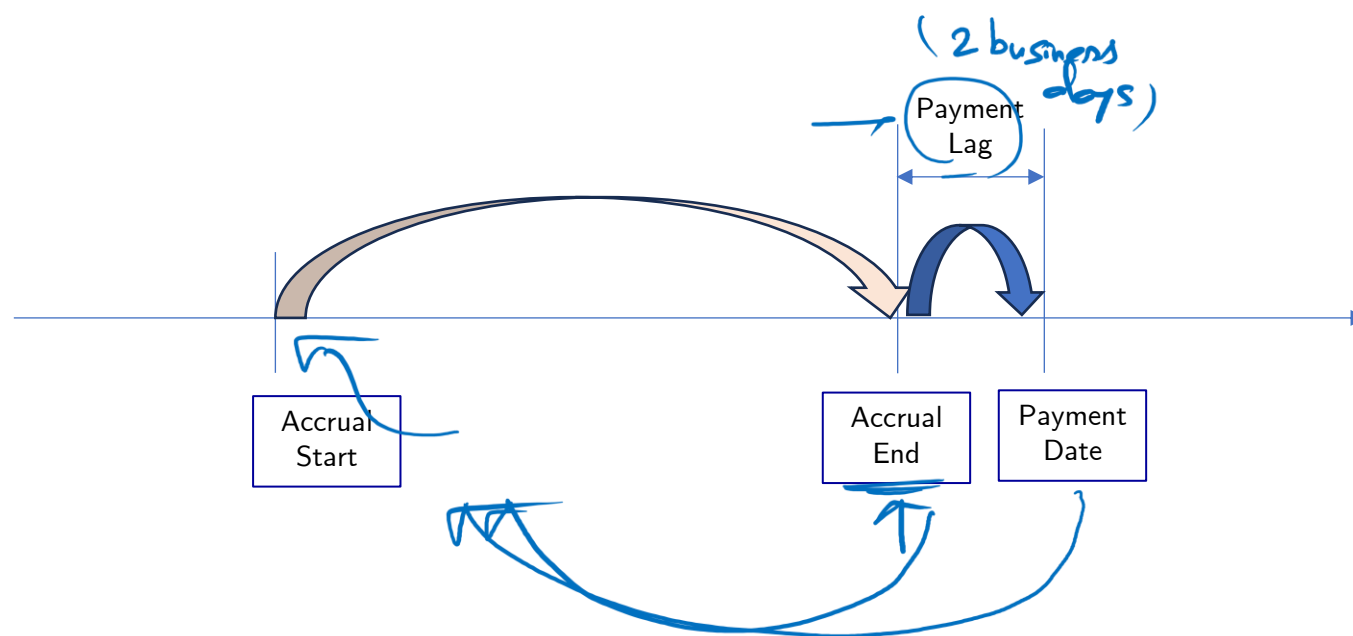
Fixing Date

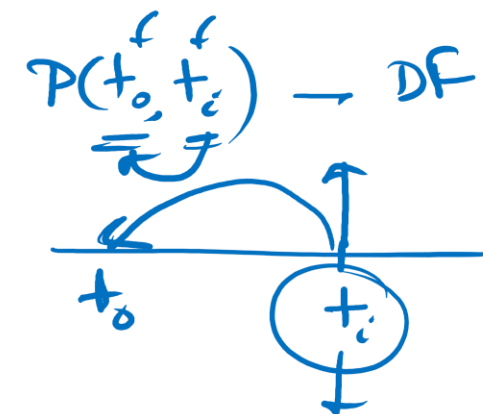
- ▶ Floating Coupon payment over an accrual period is calculated by either averaging or compounding rates over that period. But it's not necessary to use the rates observed exactly during this period. Typically, rates used in the calculation are borrowed from a few days prior to for operational convenience.
- ▶ In LIBOR days the floating rate was fixed 2 days prior to the start of the period. In RFR regime there are multiple conventions available. (More on later slides)



Payment Lag

- ▶ A **payment lag** simply postpones the payment after the accrual end. This is again to improve operational efficiency. But more importantly, the discount factor for the present value calculation need to consider the payment lag days. Typically, it's 2 business days post holiday-adjusted accrual end date.

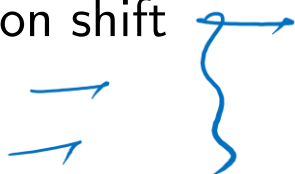




Handwritten diagram illustrating the discount factor calculation. It shows a horizontal timeline with two points, t_0 and t_i , marked by vertical lines. A blue curved arrow points from t_i back to t_0 . Above the arrow, the text $P(t_0, t_i) \rightarrow DF$ is written. A blue circle is drawn around the t_i point.

ISDA – RFR based Floating rate standard

- ▶ Given the backward-looking nature of RFR, the simple settlement in arrears standard is not practical as the full rate becomes known only hours before the payment. To overcome this practical issue, ISDA proposed a modular approach using overnight RFR rates for compounding/averaging

- ☐ Observation shift
 - ☐ Lookback
 - ☐ Lockout
- 

ISDA – RFR based Floating rate standard

	(Accrual Period 1)						(Accrual Period 2)	
	Day 1	Day 2	...	Day (T-2)	Day (T-1)	Day T	Day (T+1)	Day (T+2)
<u>Publication</u>		SOFR for Day 1	...	SOFR for Day (T-3)	SOFR for Day (T-2)	SOFR for Day (T-1)	SOFR for Day (T)	
<u>Plain Arrears</u>	Use SOFR for Day 1	Use SOFR for Day 2	...	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Use SOFR for Day (T)	Payment Due	
<u>Arrears with Payment Delay</u>	Use SOFR for Day 1	Use SOFR for Day 2	...	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Use SOFR for Day (T)		Payment Due
<u>Arrears with 1-day Lockout</u>	Use SOFR for Day 1	Use SOFR for Day 2	...	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Use SOFR for Day (T)	Payment Due	
<u>Arrears with 1-day Lookback</u>	Use SOFR for Day 0	Use SOFR for Day 1	...	Use SOFR for Day (T-3)	Use SOFR for Day (T-2)	Use SOFR for Day (T-1)	Payment Due	

Handwritten annotations:

- Arrows and circles above the table indicate accrual periods and interest rates r_1 , r_2 , and r_T .
- Under "Plain Arrears", "Payment Due" is circled and labeled "not booked".
- Under "Arrears with Payment Delay", "Payment Due" is circled and labeled "payment lag = 2".
- Under "Arrears with 1-day Lockout", "Payment Due" is circled.
- Under "Arrears with 1-day Lookback", "Payment Due" is circled.
- A red box highlights the "Use SOFR for Day (T-1)" cells in the "Arrears with 1-day Lockout" and "Arrears with 1-day Lookback" rows.
- Arrows and circles below the table indicate interest rates r_0 , r_1 , and r_{T-1} .

Lookback variants

- ▶ Let's say there is a rate r_i in the accrual period which needs to be included in the compounding and the applicable # of calendar days for r_i is d_i . Under Lookback, we use a rate observed k-days prior i.e., r_{i-k} . Lookback period is typically between 2-5 days. But there are couple of important ways in which lookback can be enforced. ↪
- ▶ **Lookback without observation shift** – We borrow only the past rate r_{i-k} but NOT the day count from the past. The applicable days for the rate to be used in compounding would be still d_i
- ▶ **Lookback with observation shift** – We borrow both the past rate, and it's associated calendar days to use in compounding.

Lookback variants

$$\left(1 + r_7 \times \frac{1}{360}\right) \left(1 + r_2\right) \quad (\text{Example : 5-days lookback})$$

	Date	d_i	SOFR
1	Mon, Jun 24, 2019	1	2.39
2	Tue, Jun 25, 2019	1	2.41
3	Wed, Jun 26, 2019	1	2.43
4	Thu, Jun 27, 2019	1	2.42
5	Fri, Jun 28, 2019	3	2.5
6	Mon, Jul 1, 2019	1	2.42
7	Tue, Jul 2, 2019	1	2.51
8	Wed, Jul 3, 2019	2	2.56
9	Fri, Jul 5, 2019	3	2.59

(without observation shift)

	Date	d_i	SOFR
	Mon, Jun 24, 2019	1	2.39
	Tue, Jun 25, 2019	1	2.41
	Wed, Jun 26, 2019	1	2.43
	Thu, Jun 27, 2019	1	2.42
	Fri, Jun 28, 2019	3	2.5
	Mon, Jul 1, 2019	1	2.42
	Tue, Jul 2, 2019	1	2.51
	Wed, Jul 3, 2019	2	2.56
	Fri, Jul 5, 2019	3	2.59

(with observation shift)

5 day
lookback

$$\left(1 + \frac{r_8 \times 2}{360}\right) \left(1 + \frac{r_3 \times 2}{360}\right)$$

Swap Schedule generation

